# Co-Op Support Application

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## I. INTRODUCTION

This project involves the development of a small-scale co-op support application designed to assist in managing the co-op admission and reporting process.

The vision of this system is to provide a simple, centralized platform that streamlines interaction between students, coordinators, and employers throughout the co-op lifecycle. The primary goal is to improve efficiency in handling applications, tracking student progress, and collecting required reports and evaluations.

The application supports three main user types: students, employers, and coordinators. Each user is required to create an account using role-specific information:

- Students: Full Name, Student ID, Email, Password
- Employers: Company Name, Supervisor Name, Password
- Coordinators: Email, Password

For Students, the application must provide the following functionality:

- Submit a PDF application
- Check application status (Pending, Accepted, Denied)
- Download work term report template
- Submit a PDF work term report

For Employers, the application must provide the following functionality:

- Submit evaluations for students based on Student ID and in the form of either:
  - PDF
  - Online form with the following fields: Behaviour, Skills, Knowledge, and Attitude

For Coordinators, the application must provide the following functionality:

- View student applications
  - Accept/reject 1st round student applications
  - Send 1st time accepted applicants to a 2nd round of screening
- View student work term reports
  - Send automated reminders to students who have not submitted their reports • View employer reports on their co-op students
- View complete list of applicants, including those accepted or rejected for both provisional and final acceptance

Overall, the system is designed to be a simple, functional, and secure, while supporting the core workflow of the co-op process.

## II. CUSTOMER REQUIREMENTS (C-REQ):

This section lists the client's requirements, each linked to a system feature.

| Req ID | Requirement | Feat. ID | Feature |
|---|---|---|---|
| C-01 | Student ID must follow a defined format (e.g. 8 digits) and be validated client-side before submission. | F1 | Application Registration |
| C-02 | Each student ID may only be associated with one active application at a time. | F1 | Application Registration |
| C-03 | All applicant data must be stored in a persistent database and not lost on server restart. | F1 | Application Registration |
| C-04 | Status change timestamps must be recorded. | F1 | Application Registration |
| C-05 | Passwords must be a minimum of 8 characters and contain at least one number and one letter. | F1 | Application Registration |
| C-06 | Must be different logins for Students, Coordinators, and Employers respectively. | F2 | Authentication & Account Management |
| C-07 | Passwords must be hashed before storage; plaintext passwords must never be stored. | F2 | Authentication & Account Management |
| C-08 | User must be kicked out of the session after 30 minutes of inactivity. | F2 | Authentication & Account Management |
| C-09 | After 5 consecutive failed login attempts, the account must be temporarily locked for 15 minutes. | F2 | Authentication & Account Management |
| C-10 | Only provisionally or finally accepted students may create a student portal account. | F2 | Authentication & Account Management |

| | | | |
|---|---|---|---|
| C-11 | Uploaded report files must be in PDF format only, reject other formats. | F3 | Work-Term Report Submission |
| C-12 | Maximum file upload size is 10 MB. | F3 | Work-Term Report Submission |
| C-13 | Submission deadline must be enforced; uploads after the deadline must be flagged or blocked. | F3 | Work-Term Report Submission |
| C-14 | Each student may only have one active report submission per work term, re-uploads must replace the previous file. | F3 | Work-Term Report Submission |
| C-15 | Uploaded PDF's must be stored securely and accessible only to the submitting student and the coordinator. | F3 | Work-Term Report Submission |
| C-16 | Employer accounts must require approval by a Coordinator before gaining access to evaluation features. | F2 | Authentication & Account Management |
| C-17 | The system must allow Coordinators to send deadline reminder notifications to students. | F3 | Work-Term Report Submission |
| C-18 | Employer evaluations must be submitted through the system and stored securely, linked to the appropriate student record. | F4 | Employer Evaluation |
| C-19 | Supervisor accounts must be created with a valid email address and verified before they can submit an evaluation | F4 | Employer Evaluation |
| C-20 | Evaluation data must be linked to the correct student and work term at the time of submission | F4 | Employer Evaluation |
| C-21 | Employers may only submit evaluations for students assigned to their organization | F4 | Employer |

| | | | |
|---|---|---|---|
| C-22 | Once submitted, evaluations must be read-only and cannot be modified by employers | F4 | Employer Evaluation |
| C-23 | A placement rejection must be recorded with a date and an optional reason field | F5 | Placement Tracking & Rejection Handling |
| C-24 | The system must distinguish between program rejection (by the coordinator) and placement rejection (by the employer) | F5 | Placement Tracking & Rejection Handling |
| C-25 | Each application must have a clearly defined final state (accepted, rejected, withdrawn) and cannot transition after finalization | F5 | Placement Tracking & Rejection Handling |
| C-26 | All rejection and acceptance actions must be logged with the acting user's role and timestamp | F5 | Placement Tracking & Rejection Handling |

## III. PRODUCT BACKLOG

This Product Backlog was developed following Scrum principles, with requirements structured as user stories and prioritized based on client needs.

### a. Personas

This defines the primary users of the system and their goals to guide the development of relevant features.

| Name | Details | Goal |
|---|---|---|
| **Alex (Student)** | A university student applying for co-op and submitting reports | "I want to apply and submit reports" |
| **Mr. Doe (Coordinator)** | Reviews applications and manages the co-op process | "I want to manage applications and track submissions" |
| **Ms. Jones (Employer)** | Supervises students and submits evaluations | "I want to evaluate students" |
| CreativeCanvasShop. (n.d.). *Images* [Photos]. Pixabay. https://pixabay.com/users/creativecanvasshop-45016655/ | | |

**b. Scenario:**

Alex (Student) fills out the co-op application form with their persona details and submits it. After being accepted into the program, Alex logs into the system and uploads a PDF report for his completed work term.

**c. Use Case:** Submit Co-op Application (Core feature)
The use case describes the system behaviour and interaction flow for a key feature.

| | |
|---|---|
| **Primary Actor** | Alex (Student) |
| **Trigger** | Alex chooses to apply for co-op |
| **Preconditions** | Alex has access to the application form |
| **Main Flow** | 1. Student opens application form<br>2. enters name, student ID, and email<br>3. Student submits the application<br>4. System validates input<br>5. System stores application<br>6. System displays confirmation message |
| **Alternative Flow** | • If required fields are missing → system shows error message<br>• If invalid data → system rejects submission |
| **Postconditions** | • Success: Application is saved in the system<br>• Failure: Application is not saved |

**d. User Stories**
User stories were written from the perspective of different system users (Students, Coordinators, Employers). The client assigned priorities, while the development team estimated effort in approximate hours. Estimates included a margin of uncertainty to account for variability in implementation and development conditions.

| ID | Related Feature / C-Req | As a… | I want to be able to… | So that… | Priority | Effort ± Hours | Status |
|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|
| S1 | F1 / C-01, C-02, C-03, C-04 | Student | Apply and submit my application for co-op | I can be considered for the program | High | 8h ± 2h | Done |
| S2 | F3 / C-11, C-12, C-13, C-14, C-15 | Student | Upload my work-term report as a PDF | I can submit my work and is stored securely | High | 1h ± 1h | In Progress |
| S3 | F2 / C-06, C-07, C-08, C-09 | Student | Log in with my credentials | I can access my account in the portal | High | 7h ± 2h | In Progress |
| S4 | F2 / C-10 | Student | I want to create a portal account after being accepted | I can submit reports | High | 5h ± 2h | Done |
| S5 | F3 / C-04 | Employer | Submit an evaluation | I can assess the student's performance | High | 6h ± 2h | Not started |
| S6 | F2 / C-06, C-07, C-08, C-09 | Coordinator | Log in securely | I can access coordinator features | High | 8h ± 2h | In Progress |
| S7 | F1 / C-03, C-04 | Coordinator | View all applications | I can review applicants | High | 6h ± 2h | Done |
| S8 | F1 / C-04 | Coordinator | Accept or reject 1st | I can manage | High | 5h ± 2h | Done |

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|---|---|---|---|---|---|---|---|
| | | | round of application s | admissions | | | |
| S9 | F1 / C-04 | Coordinator | Make final acceptance decisions later (2nd round of application s) | Requirements can be rechecked | High | 6h ± 2h | Done |
| S10 | F3 / C-13 | System | Enforce submission deadlines | Reports are submitted on time | Medium | 6h ± 2h | Done |
| S11 | F2 / C-06, C-07, C09, C-16 | Employer | Create an account | I can request access and submit evaluations | Medium | 6h ± 2h | Done |
| S12 | F3 / C-14, C-15 | Coordinator | See which students submitted reports | I can track completion | Medium | 6h ± 2h | Done |
| S13 | F3 / C-18 | Coordinator | Submit evaluations for students | They are recorded | Medium | 5h ± 1h | Not started |
| S14 | F2 / C-16 | Coordinator | Approve employer accounts | Only authorized employers can submit student evaluations | Medium | 6h ± 2h | Not started |
| S15 | F3 / C-13, C-17 | Coordinator | Send reminders | Students submit reports on time | Low | 6h ± 2h | Not started |

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|---|---|---|---|---|---|---|---|
| S16 | F2 / C-06, C-07, C-09, C-16 | Employer | Create an account | Access employer features | Low | 4h ± 1h | In Progress |
| S17 | F2 / C-05 | Student | Have my password validated for complexity (min 8 characters, at least one letter and one number) during registration | My account is secure and meets system requirements | High | 1h ± 0.5h | Done |
| S18 | F4 / C-19 | Employer | Verify my email address before submitting evaluations | Only verified supervisors can submit evaluations for students | High | 5h ± 2h | Not started |
| S19 | F4 / C-20 | Employer | Have my evaluation automaticall y linked to the correct student and work term when I submit it | Records are accurate and there's no mismatch | High | 4h ± 1h | Not started |
| S20 | F4 / C-21 | System | Restrict employers to only evaluate students assigned to their organization | Employers cannot submit evaluations for students they don't supervise | High | 5h ± 2h | Not started |

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|---|---|---|---|---|---|---|---|
| S21 | F4 / C-22 | System | Make submitted evaluations read-only for employers | Evaluation integrity is preserved after submission | Medium | 3h ± 2h | Not started |
| S22 | F5 / C-23 | Employer | Record a placement rejection with a date and an optional reason | Rejected placements are documented with context | Medium | 4h ± 1h | Not started |
| S23 | F5 / C-24 | Coordinator | Distinguish between program rejections (by coordinator) and placement rejections (by employer) | The type and source of each rejection is clear in the system | Medium | 5h ± 2h | Not started |
| S24 | F5 / C-25 | System | Lock an application into a final state (accepted, rejected, or withdrawn) with no further transitions | Application states are definitive and cannot be accidentally changed | High | 4h ± 1h | Not started |
| S25 | F5 / C-26 | System | Log all acceptance and rejection actions with the acting user's role and a timestamp | There is a full audit trail of who made each decision and when | Medium | 5h ± 2h | Not started |

**e. Story Card**
The story card expands a selected user story into implementation details, including tasks, estimates, and development notes.

***Title*: Submit co-op Application (ID: S1)**

***[requirement]***
**As a** student **I want to** apply and submit the co-op application (with personal details (name, student ID, email) **so that** I can be considered for the program

***[story]***
Application submission: Alex is presented with a form where he inputs his name, student ID, and email address. Once submitted, the system stores the application and makes it available for the co-op coordinator, Mr. Doe, to review

***[tasks]***
• Design application form UI
• Validate input fields (e.g. email format, required fields)
• Store application data in databse/list
• Create submission confirmation message

***[initial priority]***
High (set by client)

***[estimate/risk]***
8h ± 2h
Low-medium risk (basic form + validation)

***[details]***
This is a core feature of the system, as all applicants must submit their information before being reviewed. It will be one of the first features implemented in Sprint 1.

**f. Test Plan:**
For the testing phase of the co-op support application, the usage of the co-op submission form and logins with different scenarios were checked to verify the application functions accordingly. The purpose of this application is to allow students to submit co-op applications, coordinators to review and manage applications, and employers to create accounts and submit evaluations. The objective of the testing phase is aimed to ensure that features function as intended, records are stored accurately in the database, and validation and security rules are properly enforced.

Testing was conducted in a controlled test environment connected to a backend database containing the table of student applications and other user account records (including emails and hashed passwords). Manual test cases were executed using different scenarios, which include submitting applications with valid and invalid data and logging in with incorrect passwords. SQL queries were used to verify that records were correctly saved, updated, and timestamped in the database. Password lengths, file type checks, and duplicate submission checks were also tested and passed.

From the test cases table below, the results show that most features were working properly, as expected. For example, submitting an application with valid entries successfully saved the record in the database, while using invalid entries would display an error message. Viewing the list of applications worked correctly, and accepting or rejecting an application correctly updated the status and added a timestamp in the database. Attempting to submit duplicate applications ceased as well.

Authentication features were mostly tested and successful. Logging in with valid credentials worked as expected, and using invalid credentials would raise an error message. Passwords shorter than 8 characters were correctly invalidated during registration. However, account locking after 5 unsuccessful login attempts and automatic logout after 30 minutes have not yet been tested and will be validated prior to full release.

Employer account creation worked successfully, as accounts were visible in the database. File upload testing produced mixed results. Uploading invalid file types (non-PDF) uploading a report after the deadline and uploading duplicate submissions were working as expected, as all were correctly rejected. However, uploading a PDF file did not pass as the file did not correctly upload, which is an issue that needs to be addressed. Also, maximum file size validation (greater than 10 MB) was not tested, as well as evaluation submissions. Both of these scenarios will be validated before full release.

In general, the majority of application features were validated through the testing, but there were some problems that need to be fixed before release. The PDF upload failure, along with security issues such as account timeout, session locking, invalid file size check and evaluation submissions must be verified through testing and readjustments first. Once these issues are resolved, the application will be ready for release.

**g. Test Cases**

Test cases were designed to validate the functionality of key features using clear, concise steps and expected results. They are presented in a structured, table-based format for clarity and ease of evaluation.

| Test ID | Related Story ID | Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| T1 | S1 | Submit Application with valid data | Application saved in the Database, and its input (name id, email) should be queryable with:<br><br>SELECT * FROM student_application<br>WHERE student_id= id | Saved and Queryable | Passed |
| T2 | S1 | Submit Application with missing fields | An error message is displayed | A unique error message is displayed for the respective missing field | Passed |
| T3 | S7 | View list of applications | Applications are displayed | Applications are displayed | Passed |
| T4 | S8 | Accept/reject application | Status updated with timestamp in the database | Status, timestamp updated and recorded | Passed |

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|---|---|---|---|---|---|
| T5 | S6 | Log in with valid credentials | User logs in successfully | User logs in successfully | Passed |
| T6 | S2 | Upload PDF report | File uploaded successfully | File not uploaded | Failed |
| T7 | S11, S16 | Create an Employer account | Account is created, shown in the database | Shown in the database | Passed |
| T8 | S13 | Submit an evaluation | Evaluation is recoded | N/A | Not Tested |
| T9 | S6 | Login with invalid password | Error message displayed; failed attempt counted | Error message displayed; failed attempt counted | Passed |
| T10 | S6 | Login with 5 failed attempts | Account locked for 15 min | N/A | Not Tested |
| T11 | S6 | Session inactivity >30min | User logged out automatically | N/A | Not Tested |
| T12 | S2 | Upload non-PDF report | Upload rejected | Upload rejected | Passed |
| T13 | S2 | Upload report > 10MB | Upload reject | N/A | Not Tested |
| T14 | S2 | Upload the report after the deadline | Upload blocked or flagged | Upload flagged on the coordinator side | Passed |

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|---|---|---|---|---|---|
| T15 | S1 | Submit a duplicate application | Submission Blocked | Submission Blocked | Passed |
| T16 | S6 | Password is less than 8 chars or missing letters/ numbers | Registration/ login rejected | Registration & login rejected, error message displayed | Passed |
| T17 | S8, S9 | Accept/reject 1st round applicants | Move accepted applicants to 2nd round | N/A | Not Tested |
| T18 | S17 | Register with invalid input (Password less than 8 characters OR missing a number OR missing a letter) | Registration rejected with error message | N/A | Not Tested |
| T19 | S17 | Register with a valid password (8+ characters, at least one letter and one number) | Account created successfully | N/A | Not Tested |
| T20 | S18 | Employer attempts to submit an evaluation without verifying their email | Submission blocked with prompt to verify email first | N/A | Not Tested |
| T21 | S18 | Employer verifies email via verification link and then | Email marked as verified, evaluation | N/A | Not Tested |

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|---|---|---|---|---|---|
| | | submits evaluation | submission permitted | | |
| T22 | S19 | Employer submits an evaluation selecting a student and work term | Evaluation record in database is linked to the correct student ID and work term | N/A | Not Tested |
| T23 | S19 | Submit an evaluation for a selected student and work term | Returns the correct student ID and work term matching what was entered (verifying by querying the database):<br><br>SELECT student_id, work_term FROM evaluations WHERE evaluation_id = id | N/A | Not Tested |
| T24 | S20 | Employer attempts to submit an evaluation for a student not assigned to their organization | Submission blocked with error message | N/A | Not Tested |
| T25 | S20 | Employer submits an evaluation for a student assigned to their organization | Evaluation submitted and saved successfully | N/A | Not Tested |

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|---|---|---|---|---|---|
| T26 | S21 | Employer attempts to edit a previously submitted evaluation | Edit action is unavailable or blocked; evaluation remains unchanged | N/A | Not Tested |
| T27 | S21 | Coordinator views a submitted evaluation | Evaluation is displayed as read-only with all original data intact | N/A | Not Tested |
| T28 | S22 | Employer records a placement rejection with a date and a reason | Rejection saved in database with date and reason fields populated | N/A | Not Tested |
| T29 | S22 | Employer records a placement rejection with a date but no reason (optional field left blank) | Rejection saved in database with date populated and reason field null/empty | N/A | Not Tested |
| T30 | S23 | Coordinator rejects a student application (program rejection) | Rejection recorded with type "program" and coordinator's role | N/A | Not Tested |
| T31 | S23 | Employer rejects a student placement (placement rejection) | Rejection recorded with type "placement" and employer's role | N/A | Not Tested |

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|---|---|---|---|---|---|
| T32 | S23 | View a student record that has both program and placement rejections | Displays both rejection types clearly, distinguishing between “program” and “placement” along with the acting role (verifying by querying the database):<br><br>SELECT rejection_type, acting_role FROM rejections WHERE student_id = id | N/A | Not Tested |
| T33 | S24 | Coordinator attempts to change the status of an application already marked as "accepted", “rejected”, or “withdraw” (finalized) | Status change blocked; application remains in final state | N/A | Not Tested |
| T34 | S25 | Coordinator accepts a student application | Audit log entry created with coordinator role, action “accept”, and timestamp | N/A | Not Tested |
| T35 | S25 | Employer records a placement rejection | Audit log entry created with employer role, action "reject," and timestamp | N/A | Not Tested |
| T36 | S25 | View the audit log after multiple | Displays all logged actions (accept/reject) | N/A | Not Tested |

| | | | | | |
|---|---|---|---|---|---|
| | | acceptance and rejection actions | with correct acting roles and timestamps for the selected student (verified by querying the database):<br><br>SELECT acting_role, action, timestamp FROM audit_lock WHERE student_id=id | | |

## IV. Sprint 1 Backlog

Sprint 1 Backlog contains all combined user stories and their associated tasks, representing the core functionality targeted for the first sprint. This sprint focuses on implementing foundational and high-priority features, while remaining or more complex functionality will be completed in Sprint 2.

| User Story | Not Started | In Process | To Verify | Done |
|---|---|---|---|---|
| S1 **Submit Application** | — | — | — | TASK 1 **Design Application Form** |
| S15 **Send Reminders** | | | | TASK 2 **Reject and Accept student's application, with timestamp** |
| S2 **Send Reminders** | — | TASK 3 **Rejecting files that do not have a .PDF extension** | — | TASK 4 **Implement basic UI form** |
| | | TASK 12 **Evaluation PDF file submission** | | |
| S3 **Login Securely (Student)** | TASK 5 **Lockout and session timeout** | TASK 6 **Implementing UI to create a new coordinator account** | — | TASK 7 **Database and account authentication/ validation** |

| | | | | |
|---|---|---|---|---|
| S6<br>**Login Securely (Coordinator)** | | **instead of internally implementing** | | TASK 8<br>**Database successfully storing student student info securely, using hash functions to encode** |
| S4<br>**Create Portal Account after Acceptance** | – | – | – | TASK 9<br>**Student's account is automatically made once accepted** |
| S5<br>**Submit Evaluations (Employer)** | TASK 10<br>**Create a UI to write an evaluation on the website instead of uploading a file** | TASK 11<br>**Database implementation to store student evaluations** | – | – |
| S13<br>**Submit Evaluations (Coordinator)** | | TASK 12<br>**Evaluation PDF file submission** | | |
| S7<br>**View applications** | TASK 13<br>**Review and testing** | – | – | TASK 2<br>**Reject and Accept student's application, with timestamp** |
| S8<br>**Accept/ Reject Applications** | | | | |
| S9<br>**Accept Applications** | | | | |

| | | | | |
|---|---|---|---|---|
| S10 **Enforce Submission deadlines (System)** | **Not Started** | — | — | — |